ELASTIC OBSERVABILITY



Beyond the Query

Monitoring Elasticsearch-Powered Search with Elastic Observability

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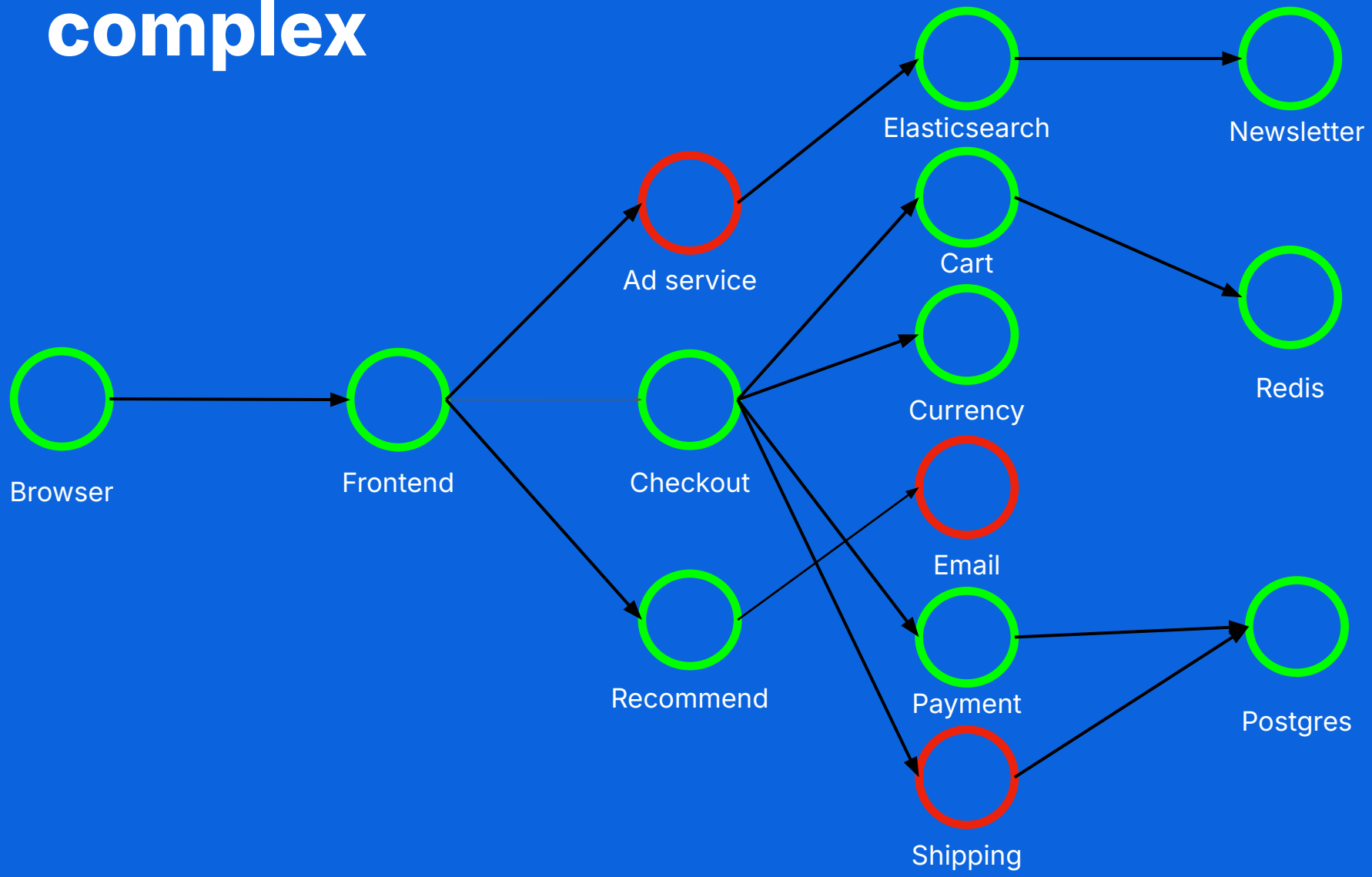
Follow-up to: "From Fluent Builders to Broken Pits" — Hans Kruse, KVK

The client ships the query.
Then what?

Hans just showed us a faster, cleaner way to talk to Elasticsearch. The moment that code hits production, a new question starts mattering more than compile-time safety: is the cluster behind it actually healthy?



Now your applications are complex



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What We’ll Cover

01



The Problem

What breaks once search goes live — and why it’s hard to see

02



What to Monitor

The signals that actually predict search trouble

03



Elastic Observability

One stack for metrics, logs, APM traces — correlated

04



The Payoff

Faster root cause, fewer fire-drills, one pane of glass

Search Looks Fine — Until It Doesn’t

Without observability, these show up as support tickets — not dashboards:

- Query latency creeps up and nobody notices until users complain
- One hot shard drags down an entire index
- Indexing falls behind during a traffic spike
- A node runs low on heap or disk, silently

THE REAL COST



Reactive firefighting

Problems surface from users first, engineers second



Guesswork debugging

Logs, metrics and traces live in separate tools



Over-provisioning

Teams add hardware instead of finding the real bottleneck

Six Signals Worth Watching



Query Latency

p50/p90/p99 response times by index and request type



Indexing Throughput

Docs/sec vs. rejected bulk requests



Cluster Health & Resources

JVM heap, CPU, disk watermarks per node



Shard Hot-Spotting

Uneven load across shards and nodes



Error & Rejection Rates

Search/index thread-pool rejections



Saturation Trends

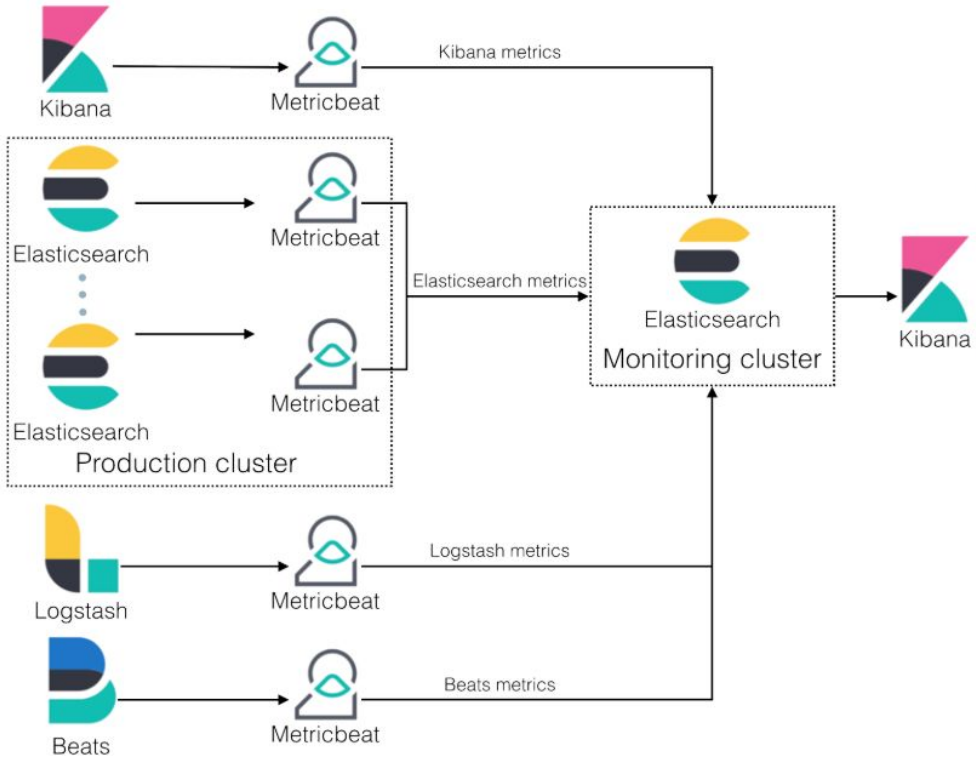
Slow, creeping degradation — not just outages

Introducing Elastic Stack monitoring

Production architecture

You can collect and ship data directly to your monitoring cluster rather than routing it through your production cluster.

The following diagram illustrates a typical monitoring architecture with separate production and monitoring clusters. This example shows Metricbeat, but you can use Elastic Agent instead.



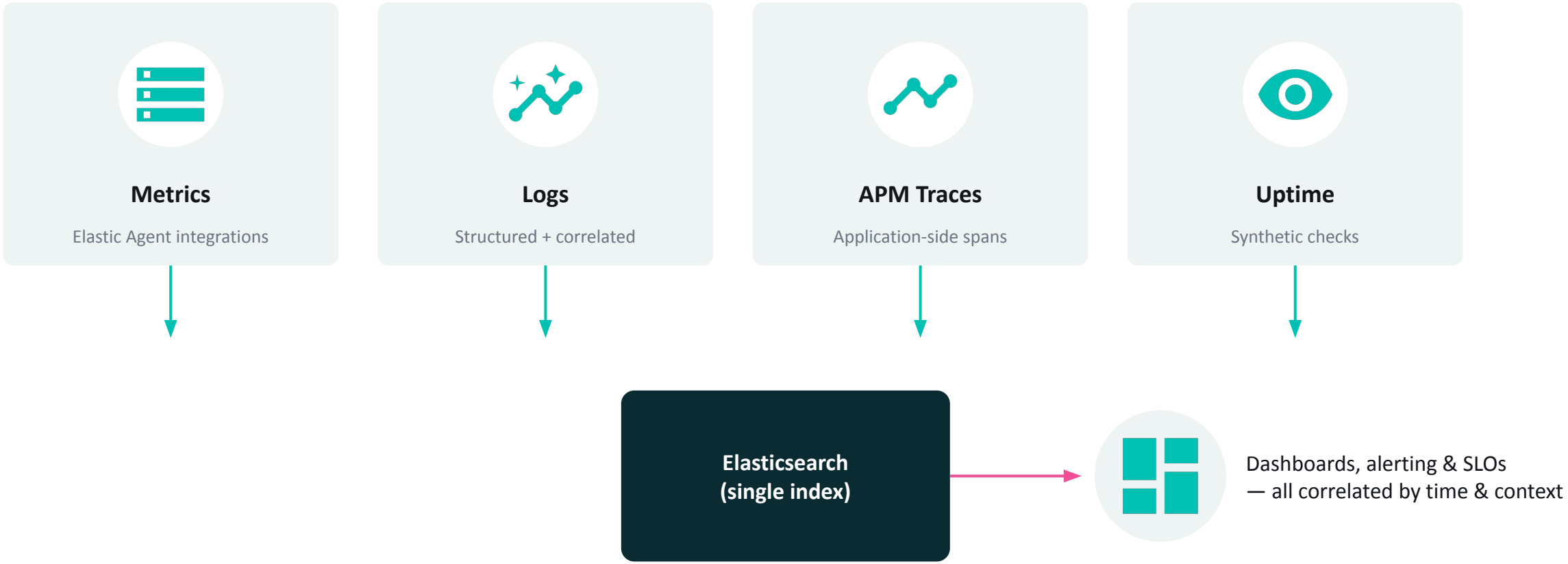
Source: https://www.elastic.co/docs/deploy-manage/monitor/stack-monitoring

Let’s watch it happen

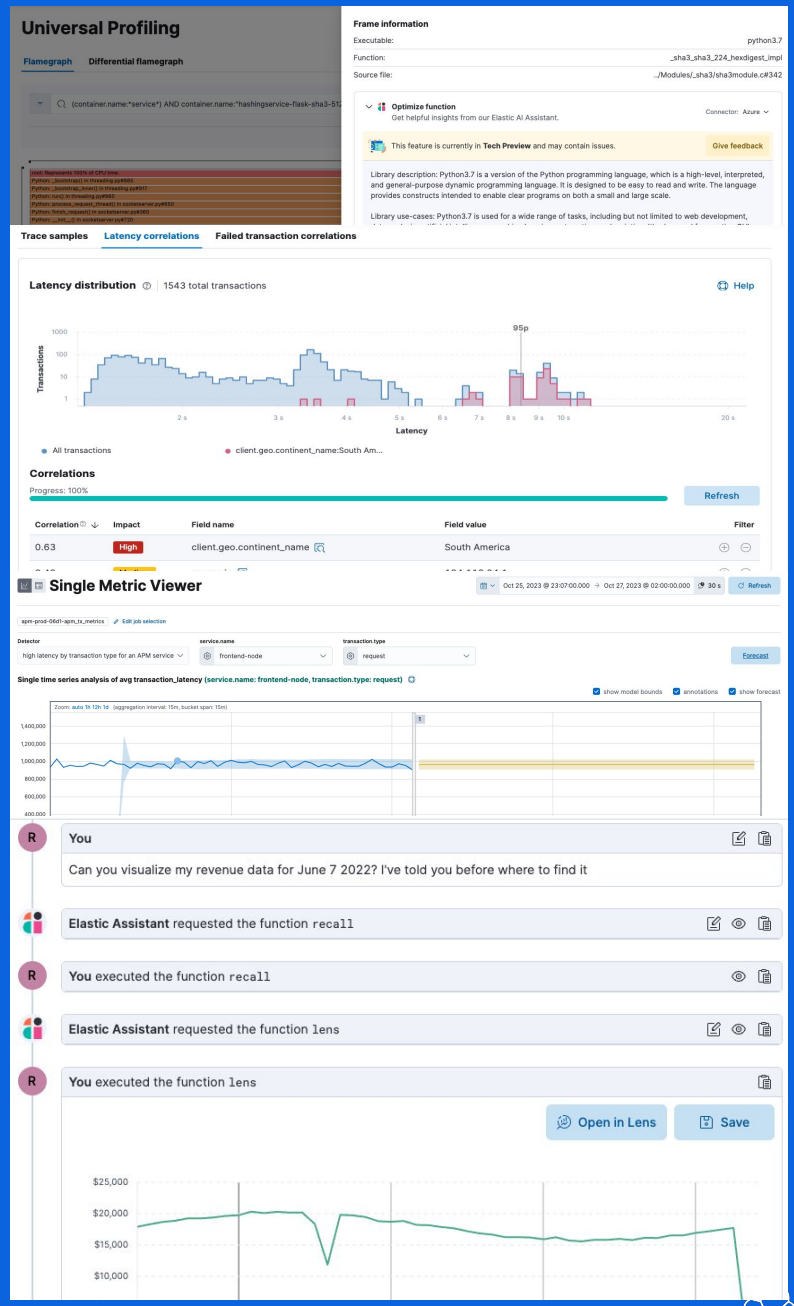
Live demo: a search-latency spike, traced from the APM span down to the hot shard behind it, with an SLO alert firing in real time.



One Stack, Not Four Tabs



Business Need	Use Cases	How to solve
Get Comprehensive Insights	Resource Optimization	Continuous Profiling
		Latency Correlations
		Historical Data Analysis
	Capacity Planning	Forecasting with Machine Learning
	Intelligent Automation	Flexible Machine Learning and Vector Search
		Autonomous AI Agents
	Proactive Incident Management	Case Management
		Context-aware AI Agents with Retrieval Augmented Generation



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From "Search Is Slow" to Root Cause

WITHOUT CORRELATION

- App team: "the API is slow"
- Search team: "the cluster looks fine"
- Separate dashboards, separate timestamps
- Hours lost reconciling two stories

WITH ELASTIC OBSERVABILITY

- APM trace shows a slow span on a search call
- One click: same trace ID in Elasticsearch cluster metrics
- Hot shard + rejected bulk requests, same moment
- Root cause found in minutes, not hours

Dashboards You Don’t Have to Build

0

custom tools to build and maintain



The Elasticsearch integration ships pre-built dashboards for cluster, node, and index health



Covers the signals from the last two slides out of the box — no custom queries needed to get started



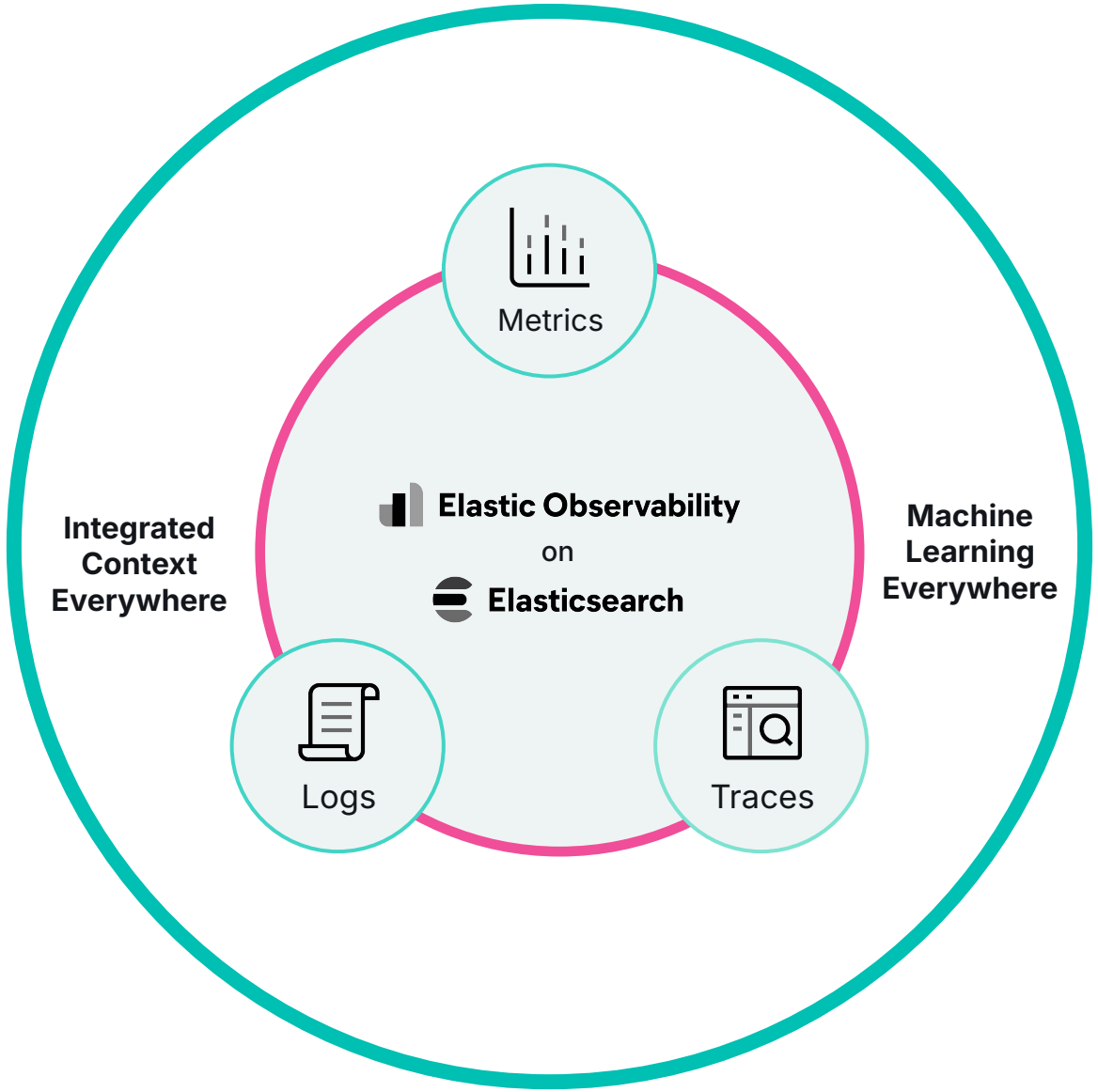
Fully customizable once the baseline is in place — extend, don’t start from scratch

Elastic advantage - One platform, single store

High cardinality, dimensionality data at scale

(keep **ALL** your data)

- **Unlimited Logging**
  no rehydration required, always indexed
- **Unlimited Tracing**
  no time limitations, no default sampling
- **Unlimited Metrics**
  no time limitations



Powerful ML & Analytics

- Anomaly detection across all telemetry
- Out-of-the box & customizable ML
- Sophisticated supervised and unsupervised ML models
- Ad-hoc analytics - understand “unknown unknowns”

From Dashboards to Proactive Alerts



Threshold Alerts

JVM heap, disk watermark, rejected requests — alert before users feel it



Anomaly Detection

ML-based baselines catch the slow creep a static threshold misses



SLO Burn-Rate Alerts

Tie alerts to a search latency SLO your stakeholders actually understand

What This Actually Buys You

Faster

root-cause analysis — minutes, not hours, correlating app + cluster

Proactive

detection — catch saturation before it’s an incident

One

pane of glass — metrics, logs and traces in the same place

Lower

tooling cost — no stitching together separate monitoring stacks

Wait a minute

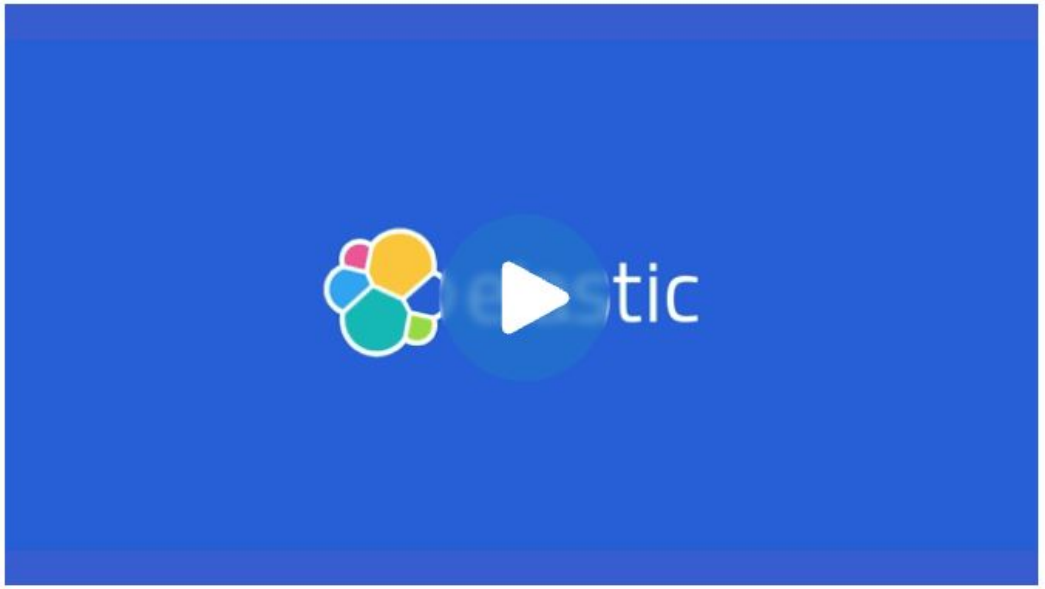
“I use Elastic free and open on-prem, what you got for me?”

Elastic AutoOps

AutoOps is now free for every Elasticsearch user

Real-time issue detection, automated root cause analysis, performance recommendations and cost insights for all self-managed clusters. No license required, no infrastructure to manage.

Connect your cluster Learn more →





Thank You

Questions?

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